

torch.linalg.tensorsolve#

<https://pytorch.org/docs/stable/generated/torch.linalg.tensorsolve.html>

Description:

Computes the solution X to the system $\text{torch.tensordot}(A, X) = B$. If m is the product of the first $B.\text{ndim}$ dimensions of A and n is the product of the rest of the dimensions, this function expects m and n to be equal. The returned tensor x satisfies $\text{tensordot}(A, x, \text{dims}=x.\text{ndim}) == B$. x has shape $A[B.\text{ndim}:]$. If dims is specified, A will be reshaped as Supports inputs of float, double, cfloat and cdouble dtypes.

Function Signature

```
torch.linalg.tensorsolve(A, B, dims=None, *, out=None) → Tensor#
```

Parameters

Keyword Arguments

Raises

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

Code Examples

```
A = movedim(A, dims, range(len(dims) - A.ndim + 1, 0))
```

```
>>> A = torch.eye(2 * 3 * 4).reshape((2 * 3, 4, 2, 3, 4))
>>> B = torch.randn(2 * 3, 4)
>>> X = torch.linalg.tensorsolve(A, B)
>>> X.shape
torch.Size([2, 3, 4])
>>> torch.allclose(torch.tensordot(A, X, dims=X.ndim), B)
True
```

```
>>> A = torch.randn(6, 4, 4, 3, 2)
>>> B = torch.randn(4, 3, 2)
>>> X = torch.linalg.tensorsolve(A, B, dims=(0, 2))
>>> X.shape
torch.Size([6, 4])
>>> A = A.permute(1, 3, 4, 0, 2)
>>> A.shape[B.ndim:]
torch.Size([6, 4])
>>> torch.allclose(torch.tensordot(A, X, dims=X.ndim), B,
atol=1e-6)
True
```

Notes & Warnings

NOTE

See also `torch.linalg.tensorinv()` computes the multiplicative inverse of `torch.tensordot()`.

Detailed Documentation

Documentation

Computes the solution X to the system $\text{torch.tensordot}(A, X) = B$.

If m is the product of the first $B.\text{ndim}$ dimensions of A and n is the product of the rest of the dimensions, this function expects m and n to be equal.

The returned tensor x satisfies $\text{tensordot}(A, x, \text{dims}=x.\text{ndim}) == B$. x has shape $A[B.\text{ndim}:]$.

If dims is specified, A will be reshaped as

Supports inputs of float, double, cfloat and cdouble dtypes.

`torch.linalg.tensorinv()` computes the multiplicative inverse of `torch.tensordot()`.

A (Tensor) – tensor to solve for. Its shape must satisfy $\text{prod}(A.\text{shape}[:B.\text{ndim}]) == \text{prod}(A.\text{shape}[B.\text{ndim}:])$.

B (Tensor) – tensor of shape $A.\text{shape}[B.\text{ndim}]$.

dims (Tuple[int], optional) – dimensions of A to be moved. If None, no dimensions are moved. Default: None.

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

RuntimeError – if the reshaped $A.\text{view}(m, m)$ with m as above is not invertible or the product of the first ind dimensions is not equal to the product of the rest of the dimensions.

